

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19- 304 - 1

MINIATURE CIRCUIT BREAKERS (MCBs)

Date : 3-3- 2026

EDMS 19-304-1
SPECIFICATION
FOR
MINIATURE CIRCUIT BREAKERS (MCBs)

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 19- 304 - 1****MINIATURE CIRCUIT BREAKERS (MCBs)****Date : 3-3- 2026****1. SCOPE**

This specification covers the minimum technical requirements for design, manufacturing, testing, inspection, packing and delivery of Miniature Circuit Breakers (MCBs) intended for use in:

- Meter enclosures, and
- All low voltage applications of Electricity Distribution Companies (EDCs)

The MCBs shall be suitable for indoor installation in low voltage systems, panels and meter enclosures operating (single / three phase direct smart meters) under normal service conditions of the LV distribution network.

2. APPLICABLE STANDARDS

No.	Standard	Description
1	IEC 60898-1	Circuit breakers for overcurrent protection for household and similar installations
2	IEC 60947-2	Low-voltage switchgear and control gear – Circuit breakers (where applicable)
3	IEC 60664	Insulation coordination for equipment within low-voltage systems
4	IEC 60529	Degrees of protection provided by enclosures (IP Code)
5	IEC 60695-2	Fire hazard testing – Part 2-11 : Glowing/hot-wire based test – methods Glow- wire flammability test method for end-products

3. ENVIRONMENTAL CONDITIONS

Item	Condition
Ambient temperature	-5 °C to +40 °C
Maximum relative humidity	95 %
Altitude	Up to 1000 m
Installation	Indoor
Pollution degree	Degree 2 (IEC 60664)

Any deviation from the above conditions shall be clearly stated in the offer.

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4. DESIGN CRITERIA

1. Type of Construction

The Miniature Circuit Breakers (MCBs) shall be of the **thermal-magnetic type**, providing:

- **Overload protection** by means of a thermal (bimetal) release, and
 - **Short-circuit protection** by means of a magnetic instantaneous release.
- The tripping mechanism shall operate independently of the operating handle position.

2. Mounting and Installation

- MCBs shall be provided with a **fast clip** DIN rail mounting system suitable for **35 mm²** standard **DIN** rails, allowing quick installation and removal without special tools, while ensuring secure locking and vibration resistance during operation and maintenance.
- Terminals shall be of **cage clamp type** suitable for secure connection of copper conductors with cross-sections up to 35 mm², ensuring reliable electrical contact, vibration resistance, and thermal stability.
- The design shall allow easy installation and removal without damage to adjacent devices.
- MCBs shall be suitable for use in **meter enclosures, distribution boards and modular panels**.

3. Degree of Protection and Insulation

- Terminals shall comply with a minimum degree of protection of **IP20** in accordance with IEC 60529, ensuring protection against finger contact with live parts during installation, operation, and maintenance.

4. Insulating Materials

- All insulating parts shall be manufactured from non-flammable, self-extinguishing material with flammability classification **V0** according to UL 94 or equivalent tests according to IEC 60695-11 standards.
- The materials shall be resistant to:
 - Heat generated during normal operation and fault conditions,
 - Ageing under electrical and thermal stresses, and
- The insulating materials shall comply with the requirements of **IEC 60898-1** regarding resistance to abnormal heat and fire.

5. Operating Mechanism and Indication

- MCBs shall be provided with a **positive, quick-make and quick-break operating mechanism**.
- For multi- pole circuit breakers (2P,3P ,and 4P) , all poles shall mechanically linked and shall open and close simultaneously ,tripping of any pole shall cause the opening of all others poles .

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- A **clear and durable mechanical indication** of the operating position shall be provided:

- **ON (closed)**
- **OFF (open)**

6. Tripping Performance

- Under short-circuit conditions, the MCB shall **trip instantaneously** within the magnetic tripping range corresponding to the selected curve (B, C ,D,K or Z)
- Tripping characteristics shall remain stable over the service life of the device.

7. Current-Carrying Parts

- All current-carrying components shall be made of **high-conductivity copper with a minimum purity of 99.9%**.
- Contact surfaces shall be suitably plated or treated to:
 - Minimize contact resistance,
 - Prevent corrosion, and
 - Ensure consistent performance throughout the service life.

8. Arc Quenching System

- MCBs shall be equipped with an **Conventional arc chamber with arc splitters and arc runners**.
- designed to:
 - Rapidly extinguish the arc during interruption,
 - Limit arc energy, and
 - Reduce contact erosion.

9. Endurance and Reliability

The MCBs shall demonstrate sufficient mechanical and electrical endurance to ensure reliable operation throughout their service life:

- **Mechanical endurance:** The MCB shall withstand a minimum of **20,000** operating cycles for rated currents, in accordance with IEC 60898 / IEC 60947 requirements, without deterioration in performance or safety.
- **Electrical endurance:**
 - For rated current up to 63 A: not less than 10,000 cycles at rated current (In).
 - For rated current above 63 A up to 100 A: not less than 5,000 cycles at rated current (In).
 - For rated current above 100 A: not less than 4,000 cycles at rated current (In).

10. Minimum Clearances and Creepage Distances

(For 220 / 400 V AC Systems – IEC 60898-1, Pollution Degree 2)

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Condition	Minimum Clearance (mm)	Minimum Creepage Distance (mm)
Between live parts with contacts in open position	4.0	4.0
Between live parts and earthed metal parts	3.0	3.0

2. Between Circuits Supplied from Different Sources

Item	Minimum Clearance (mm)	Minimum Creepage Distance (mm)
Circuits supplied from different sources	6.0	6.0

3. Between Live Parts and Accessible Parts

Condition	Minimum Clearance (mm)	Minimum Creepage Distance (mm)
Operating parts, covers, and accessible metal frames	3.0	3.0

11.- Maximum Power Loss per Pole (IEC 60898-1, Clause 8.13)

Rated Current Range (In)	Maximum Power Loss per Pole
$I_n \leq 10 \text{ A}$	3 W
$10 < I_n \leq 16 \text{ A}$	3.5 W
$16 < I_n \leq 25 \text{ A}$	4.5 W
$25 < I_n \leq 32 \text{ A}$	6 W
$32 < I_n \leq 40 \text{ A}$	7.5 W
$40 < I_n \leq 50 \text{ A}$	9 W
$50 < I_n \leq 63 \text{ A}$	13 W
$63 < I_n \leq 100 \text{ A}$	15 W
$100 < I_n \leq 125 \text{ A}$	20 W

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The power loss values indicated above are specified **per pole**. Compliance shall be verified in accordance with **IEC 60898-1, Clause 9.8.5**.

5. ELECTRICAL CHARACTERISTICS

Item	Requirement
Rated operational voltage (U_n)	230 / 400 V AC
Rated frequency	50 Hz
Rated current (I_n)	6, 10, 16, 20, 25, 32, 40, 50, 63, 80, 100, 125 A
Rated insulation voltage (U_i)	$\geq 250/440$ V
Rated impulse withstand voltage (U_{imp})	≥ 4 kV

6. SHORT-CIRCUIT BREAKING CAPACITY

Duty Level	Rated Breaking Capacity (I_{cn})
Minimum duty	4.5 kA
Standard LV distribution	6 kA
Heavy duty / critical points	10 kA up to 25 kA

The required breaking capacity shall be defined by the concerned **EDC** according to installation location.

7. TRIPPING CHARACTERISTICS

The required curve type shall be defined by the concerned **EDC** according to installation location.

Magnetic Tripping Characteristics – Complete Curve Types

Curve Type	Magnetic Tripping Range
Type B	$3 - 5 \times I_n$
Type C	$5 - 10 \times I_n$
Type D	$10 - 20 \times I_n$
Type K	$10 - 14 \times I_n$
Type Z	$2 - 3 \times I_n$

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Type	No. of Poles	Description	Rated Voltage
1P	1	Single phase (phase only)	230 V
2P	2	Phase + Neutral	230 V
3P	3	Three phase (without neutral)	400 V
4P	4	Three phase + Neutral	400 V

9. MCB TECHNICAL DATA TABLE

Poles	Rated Current (A)	Breaking Capacity (kA)	Ui (V)	Uimp (kV)
1P / 2P	6 – 125	4.5-25 KA	≥250/440	≥4
3P / 4P	6 – 125			≥4

10. INSPECTION AND TESTING

- EEHC / EDC reserves the right to witness routine tests during the homologation process.
- All MCBs shall be tested according to IEC 60898-1

10.1 Type Tests (According to IEC 60898-1)

The Miniature Circuit Breakers (MCBs) shall be subject to the following type tests:

- Indelibility of marking (Clause 9.3)
- Reliability of screws, current-carrying parts and connections (Clause 9.4)
- Reliability of terminals for external conductors (Clause 9.5)
- Protection against electric shock (Clause 9.6)
- Dielectric properties and isolating capability (Clause 9.7)
- Temperature-rise test (Clause 9.8)
- 28-day test (Clause 9.9)
- Tripping characteristic test (Clause 9.10)
- Mechanical and electrical endurance test (Clause 9.11)
- Short-circuit test (Clause 9.12)
- Resistance to mechanical shock and impact (Clause 9.13)
- Resistance to heat (Clause 9.14)
- Resistance to abnormal heat and to fire (Clause 9.15)

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- Resistance to rusting (Clause 9.16)

10.2 Routine Tests according to (IEC 60898-1)

The following routine tests shall be carried out on each manufactured MCB to ensure quality and conformity:

- Visual and dimensional inspection** of the MCB.
- Mechanical operation test:** check ON / OFF
- Verification of tripping performance** according to the rated current and selected curve (B, C, D ,K, Z).
- Dielectric test:** verify insulation withstand at rated voltage.
- Continuity test** of current-carrying parts and terminals.
- Verification of markings:** ensure all markings are present and legible.
- Tightness of terminal screws:** ensure correct torque application.
- Isolation function check:** confirm that open contacts provide visible and reliable isolation gap.

11. PACKING & SHIPPING

- Original sealed manufacturer packing.
- Protection against dust, moisture, mechanical damage.
- Packing shall be marked with: Manufacturer name, Country of origin, Rated current, Breaking capacity.
- The supplied MCBs shall be provided through the manufacturer's/Mother Company's official distributors or authorized partners, or supported by a Manufacturer/Mother Company Authorization Letter issued by the original manufacturer/ Mother Company, confirming that the bidder is authorized to supply the offered MCBs for this tender.
- All authorization documents shall be valid and issued by the manufacturer or the Mother Company, and the EEHC/EDC reserves the right to verify their authenticity directly with the manufacturer/the Mother Company.

12. MARKING

- All markings, symbols and ratings shall be **clear, indelible and legible** throughout the service life of the MCB.
- Markings shall remain readable after exposure to normal operating, thermal and environmental conditions.

- Manufacturer name
- Country of origin
- Rated current (In)

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4. Rated voltage (Un)
5. Tripping curve (B / C / D,K,Z)
6. Rated short-circuit breaking capacity (Icn)
7. Applicable IEC standard

13. SUBMITTALS

1. Technical catalogues and datasheets
2. Type test certificates according to IEC standards
3. Filled guarantee table
4. Approved EEHC / EDC reference letter (if required)

14. GUARANTEE

- 18 months from delivery or **12** months from operation
- Defective MCBs shall be replaced free of during the warranty period

15. OPTION LIST**15.1 Tripping Curve Options**

No.	Curve Type	Required	Not Required
1	Type B ($3-5 \times I_n$)	☐	☐
2	Type C ($5-10 \times I_n$)	☐	☐
3	Type D ($10-20 \times I_n$)	☐	☐
4	Type K ($10-14 \times I_n$)	☐	☐
5	Type Z ($2-3 \times I_n$)	☐	☐

15.2 Rated Current Options

No.	Rated Current	Required	Not Required
1	6 A	☐	☐
2	10 A	☐	☐
3	16 A	☐	☐
4	20 A	☐	☐
5	25 A	☐	☐
6	32 A	☐	☐

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No.	Rated Current	Required	Not Required
7	40 A	☐	☐
8	50 A	☐	☐
9	63 A	☐	☐
10	80 A	☐	☐
11	100 A	☐	☐
12	125 A	☐	☐

15.3 Number of Poles Options

No.	Type	Description	Required	Not Required
1	1P	Single pole	☐	☐
2	2P	Phase + Neutral	☐	☐
3	3P	Three phase	☐	☐
4	4P	Three phase + Neutral	☐	☐

15.4 Short-Circuit Breaking Capacity Options

No.	Rated Breaking Capacity (Icn)	Required	Not Required
1	4.5 kA	☐	☐
2	6 kA	☐	☐
3	10 kA	☐	☐
4	25 KA	☐	☐

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MCB Technical Data Schedule

Parameter	Description / Value	Supplier / Manufacturer
Rated Operational Voltage (Un)	230 / 400 V AC	☐
Rated Frequency	50 Hz	☐
Rated Current (In)	6, 10, 16, 20, 25, 32, 40, 50, 63, 80, 100, 125 A	☐
Insulation Voltage (Ui)	≥ 250/440 V	☐
Impulse Withstand Voltage (Uimp)	≥ 4 kV	☐
Short-Circuit Breaking Capacity (Icu / Icn)	4.5 / 6 / 10/ 25 kA	☐
Tripping Curves	Type B / C / D /k /Z	☐
Number of Poles	1P / 2P / 3P / 4P	☐
Mechanical Endurance	≥ 20,000 ON/OFF operations	☐
Electrical Endurance	For rated current up to 63 A: not less than 10,000 cycles at rated current (In). For rated current above 63 A up to 100 A: not less than 5,000 cycles at rated current (In). For rated current above 100 A: not less than 4,000 cycles at rated current (In).	☐
Degree of Protection (IP)	IP20	☐
Ambient Temperature	-5 °C to +40 °C	☐
Maximum Relative Humidity	95 %	☐
Altitude	Up to 1000 m	☐
Pollution Degree	2	☐
Mounting	DIN rail (35 mm)/ cage terminals / fast clip	☐
Insulation Type	non-flammable, self-extinguishing according to UL94 or IEC /EN 60695-11:V0	☐
ON / OFF Indication	Mechanical	☐
Materials	Copper ≥99.99%	☐
Arc Quenching System	Conventional arc chamber with arc splitters and arc runners.	☐
Power Loss per Pole	3 – 20 W	☐
Isolation Function	Yes	☐
Creepage & Clearance Distances	As per IEC 60898-1 Table 4	☐